

2025). Despite this critical impact, we lack a systematic and grounded understanding of the evolving landscape. From a coverage perspective, prior research often focuses on a few domains like software engineering (Hoffmann et al., 2024) and customer support (Brynjolfsson et al., 2025). This narrow scope limits our comprehension of the real-world complexity of diverse human jobs and the varied nature of open-ended tasks. From a stakeholder perspective, existing studies often emphasize the interests of capital by focusing on a few tasks that tend to be more profitable such as coding without adequately considering worker values (Eisfeldt et al., 2023). Furthermore, current approaches often rely on analyzing existing usage data, such as how people use chatbots for work (Hazra et al., 2025, Zhao et al., 2024), which cannot provide a forward-looking assessment of AI potential across the broader workforce.

To address these gaps, we propose a principled, survey-based framework to investigate which occupational tasks workers want AI agents to automate or augment. We look at the entire workforce that could be impacted by digital AI agents by sourcing occupational tasks from the U.S. Department of Labor’s O*NET database. Compared to occupation-level studies, task-level auditing allows us to better capture the nuanced, open-ended, and contextual nature of real-world work. Our auditing framework takes a worker-centric approach by soliciting first-hand insights from domain workers actively performing the tasks. To guide domain workers in providing well-calibrated responses, we empower them to share their experiences and articulate their reasoning through an audio-enhanced survey system. Crucially, our framework expands beyond the binary view of automation. We propose the Human Agency Scale (*i.e.*, H1-H5), which complements the SAE L0-L5 automation levels (Committee et al., 2014) by quantifying the degree of human involvement required for occupational task completion and quality. This new scale centers human agency—a crucial factor for responsible AI agent adoption (Fanni et al., 2023)—and provides a shared language to capture the spectrum between automation and augmentation.

To ground workers’ perspectives in technical reality, we further gather complementary assessments from AI experts with experience in agent research and development (R&D). This dual approach reveals how workers and experts perceive AI agents’ capabilities and risks at work. Based on data collected through January 2025 to May 2025, we construct the AI Agent Worker Outlook & Readiness Knowledge Bank (WORKBank). This database currently consists of responses from 1,500 workers across 104 occupations and annotations from 52 AI experts, covering 844 occupational tasks. It is designed to be easily extensible to more tasks and to reflect evolving technological capabilities and worker preferences. To our knowledge, this is the first large-scale audit of AI agent capabilities and worker preferences.

Our work contributes four sets of findings:

1. **Domain workers want automation for low-value and repetitive tasks (Figure 4).** For 46.1% of tasks, workers express positive attitudes toward AI agent automation, even after reflecting on potential job loss concerns and work enjoyment. The primary motivation for automation is freeing up time for high-value work, though trends vary significantly by sector.
2. **We visualize the desire-capability landscape of AI agents at work, and find critical mismatches (Figure 5).** The worker desire and technological capability divide the landscape into four zones: Automation “Green Light” Zone (high desire and capability), Automation “Red Light” Zone (high capability but low desire), R&D Opportunity Zone (high desire but currently low capability), and Low Priority Zone (low desire and low capability). Notably, 41.0% of company-task mappings are concentrated the Low Priority Zone and Automation “Red Light” Zone. Current investments mainly center around software development and business analysis, leaving many promising tasks within the “Green Light” Zone and Opportunity Zone under-addressed.
3. **The Human Agency Scale provides a shared language to audit AI use at work and reveals distinct patterns across occupations (Figure 6).** 45.2% of occupations have H3 (equal partnership)